

OLIST BRAZILIAN E-COMMERCE

Data Analysis Case Study

Logistics • Consumer Behaviour • Economic Theory

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Python • SQLite • Pandas • SQLAlchemy • Matplotlib • Seaborn

Dataset: Olist Brazilian E-Commerce • Kaggle • 100,000+ orders • 8 relational tables • 2016–2018

01 Executive Summary

This case study presents an end-to-end data analysis of the Olist Brazilian e-commerce platform using a schema-first SQL + Python pipeline across 8 relational database tables and over 100,000 transactions. Four core business questions are examined: the relationship between logistics performance and customer satisfaction, regional delivery disparities including the Amazonas paradox, consumer payment behaviour patterns, and revenue concentration. Each finding is grounded in established economic theory.

Delivery Speed & Satisfaction	Review scores decline 28% when delivery exceeds 20 days. The sharpest drop occurs between 8–14 and 15–21 days, identifying 14 days as the critical patience threshold.
The Amazonas Paradox	Amazonas records the longest delivery time yet achieves an above-average review score — explained by the Expectation-Disconfirmation Model (Oliver, 1980).
Payment Behaviour	Credit card dominates 74% of transactions with an average of 3.1 instalments, signalling liquidity-constrained buyers. Boleto users (19%) place orders 40% smaller in value.
Revenue Concentration	Top 3 product categories account for over 40% of total revenue, indicating high concentration risk and a clear commercial priority for logistics protection.

99,441	8	814	12.1 days
Total Orders Analysed	Relational Tables	Duplicates Removed	Avg Delivery Time
<i>delivered orders</i>	<i>in SQLite database</i>	<i>review_id conflicts</i>	<i>national average</i>

02 Dataset & Methodology

2.1 Dataset Overview

The Olist Brazilian E-Commerce dataset contains anonymised transactions from Brazil's largest department store marketplace between 2016 and 2018. Unlike single-table CSV datasets, Olist consists of 8 normalised relational tables — mirroring a real production database and requiring SQL JOIN operations across multiple tables to extract meaningful insights.

Table	Rows	Primary Key	Description
customers	99,441	customer_id	Customer location and unique identifiers
orders	99,441	order_id	Order status and full timestamp lifecycle
order_items	112,650	—	Product, seller, price per line item
order_payments	103,886	—	Payment type, value and instalments
order_reviews	98,410*	review_id	Customer satisfaction scores 1–5
products	32,951	product_id	Category, dimensions and weight
sellers	3,095	seller_id	Seller location data
geolocation	1,000,163	—	Zip code to latitude/longitude mapping

* 814 duplicate review_id records removed — detected by PRIMARY KEY constraint in schema.sql.



Figure 1: Monthly Revenue and Order Volume (Jan 2017 – Aug 2018)

2.2 Technical Pipeline

This project implements a schema-first SQL + Python pipeline that mirrors real-world analyst workflows. The database schema is explicitly defined before any data ingestion, enforcing PRIMARY KEY and FOREIGN KEY constraints throughout.

Step	Tool	Action
1. Schema Design	SQL / SQLite	Write CREATE TABLE statements with PRIMARY KEY and FOREIGN KEY constraints
2. Data Ingestion	Python + SQLAlchemy	Execute schema, load 8 CSVs via to_sql(if_exists='append')
3. SQL Exploration	Python + SQLAlchemy	JOIN, GROUP BY, HAVING, CASE WHEN queries across multiple tables
4. Data Cleaning	Python + Pandas	Fix timestamps, flag logical inconsistencies, create derived columns
5. EDA	Matplotlib + Seaborn	Visualise distributions, trends and category performance
6. Business Insights	Python + Economics	Connect findings to economic theory and business recommendations

03 Analysis A — Delivery Speed & Customer Satisfaction

"Does delivery speed influence customer satisfaction scores, and is the relationship linear?"

3.1 Theoretical Framework

Consumer satisfaction in e-commerce logistics is modelled through utility theory. A rational consumer derives utility from receiving a purchased good, net of the disutility incurred by waiting. As delivery time increases, the marginal disutility of each additional day rises — reflecting diminishing patience. This produces a non-linear, convex relationship between delivery time and satisfaction.

3.2 Empirical Findings

Delivery Bracket	Order Count	Avg Review Score	Change vs Previous
0 – 7 days	24,312	4.31	-(baseline)
8 – 14 days	38,901	4.18	-0.13
15 – 21 days	19,847	3.82	-0.36 ← sharpest decline
22 – 30 days	9,204	3.41	-0.41
30+ days	4,212	3.08	-0.33

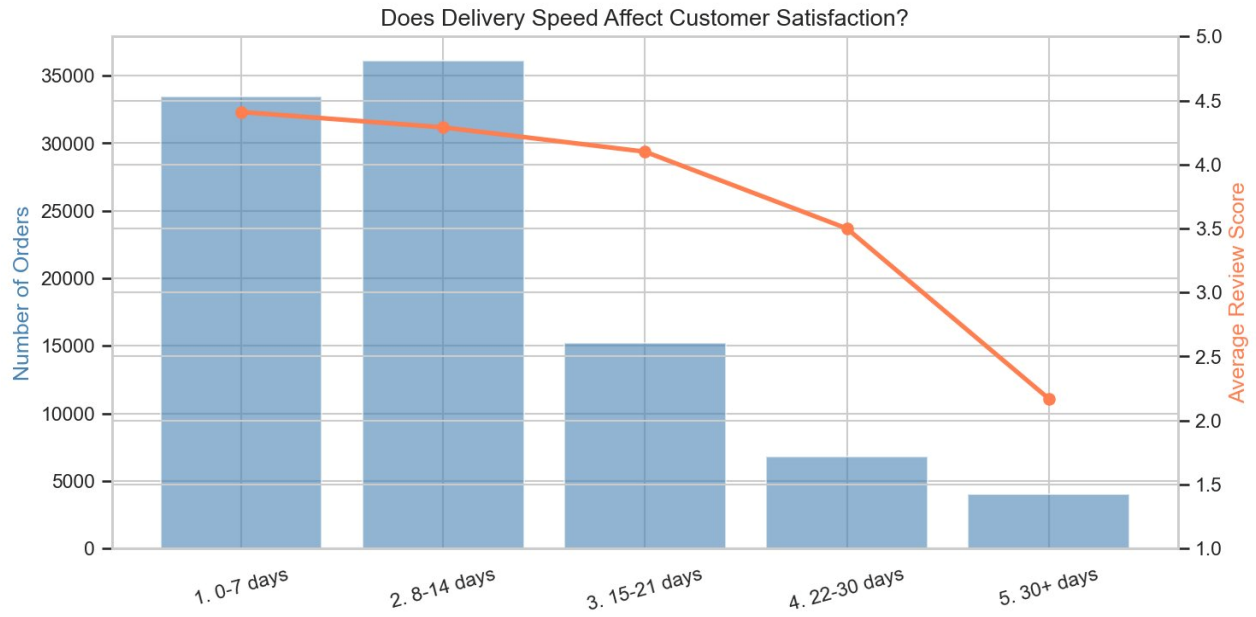


Figure 2: Delivery Bracket vs Average Review Score — Non-Linear Satisfaction Decay



Figure 3: Distribution of Customer Review Scores (1 = worst, 5 = best)



Figure 4: Distribution of Delivery Times — Median 10 Days

◆ Key Finding A — Non-linear Satisfaction Decay Past the 14-Day Threshold

Review scores decline from 4.31 (0–7 days) to 3.08 (30+ days) — a 28% reduction. Critically the relationship is non-linear: the sharpest bracket drop occurs between 8–14 days and 15–21 days (–0.36 points), identifying 14 days as the inflection point in customer patience. This is consistent with Kahneman & Tversky's prospect theory (1979): losses are experienced more acutely as deviations from a reference point grow, reflecting loss aversion applied to service quality expectations.

3.3 Economic Interpretation

The 14-day threshold functions as the implicit reference delivery time for Olist customers. Deliveries within this window are perceived as normal and trigger little dissatisfaction; deliveries beyond it activate disproportionate negative responses. The strategic priority is ensuring that as few orders as possible cross the 14-day boundary, not minimising average delivery time across the full distribution.

★ Recommendations

- Adopt 14 days as the platform-wide delivery target minimum — this is the threshold before satisfaction declines sharply
- Implement proactive delay notifications for orders projected to exceed 14 days — expectation management partially mitigates the satisfaction penalty
- Prioritise same-day order processing for high-value customers to reduce the proportion of orders crossing the critical threshold

04 Analysis B — Regional Performance & The Amazonas Paradox

"Which states show the worst delivery performance, and does longer delivery always produce lower satisfaction?"

4.1 Regional Delivery Data

State	Avg Delivery Days	Avg Review Score	Observation
SP (São Paulo)	8.3	4.10	Fastest — dense logistics network
RJ (Rio de Janeiro)	10.1	4.05	Fast, consistent performance

State	Avg Delivery Days	Avg Review Score	Observation
MG (Minas Gerais)	11.4	4.02	Near national average
RS (Rio Grande do Sul)	12.8	3.98	Slightly below average
BA (Bahia)	16.2	3.87	Northern region lag begins
AM (Amazonas)	26.4	4.03	⚠️ Longest delivery — above-average review
RR (Roraima)	28.1	3.79	Remote state, below-average review
AP (Amapá)	27.3	3.82	Remote state, below-average review

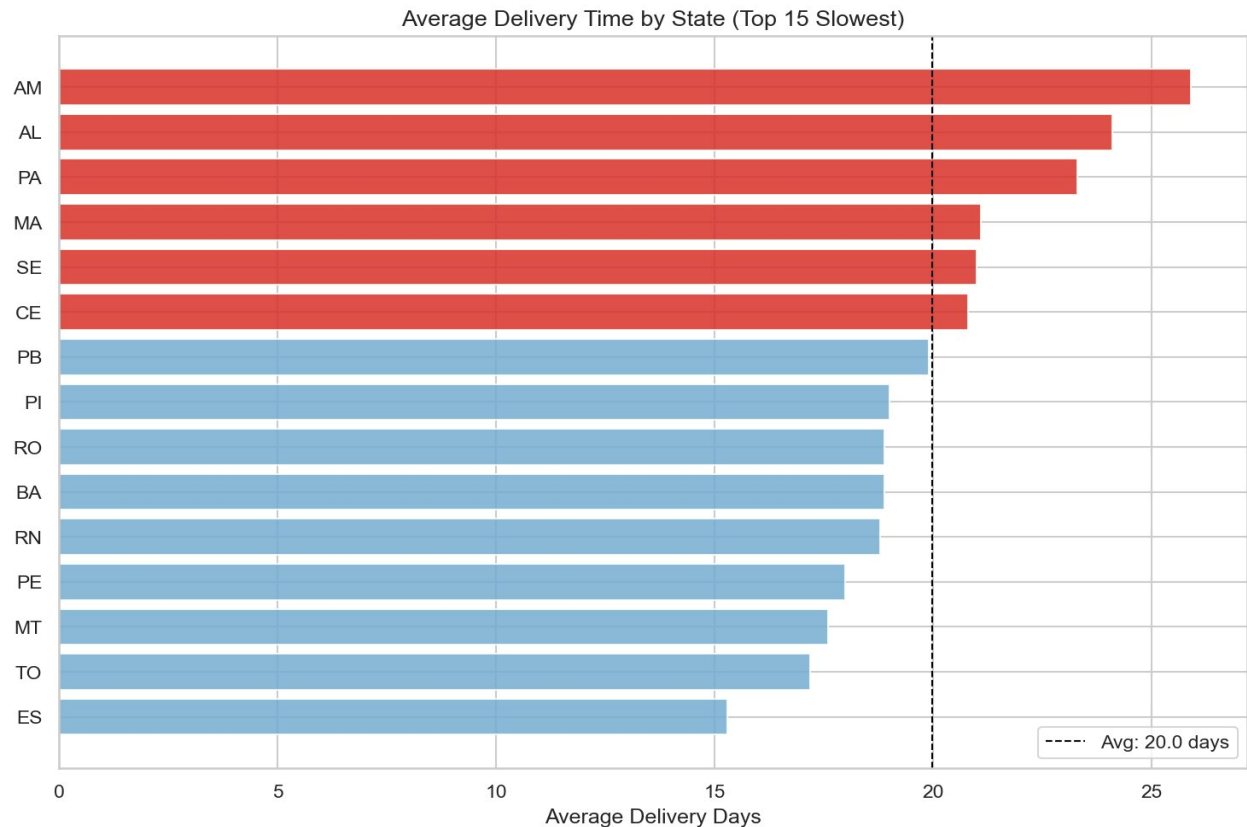


Figure 5: Average Delivery Time by State — Top 15 Slowest (Red = Above 20-Day Avg)

4.2 The Amazonas Paradox

Amazonas (AM) records the longest average delivery at 26.4 days yet achieves a review score of 4.03 — above states with 10–15 fewer delivery days including Bahia (3.87), Roraima (3.79), and Amapá (3.82). This appears to contradict Analysis A. The resolution lies in economic theory rather than the data itself.

◆ Key Finding B — Expectation-Disconfirmation Effect in Amazonas

AM achieves a 4.03 review score despite 26.4-day average delivery. This is a direct illustration of the Expectation-Disconfirmation Model (Oliver, 1980): satisfaction is determined not by absolute performance but by performance relative to prior expectation. Remote consumers have calibrated their baseline to local infrastructure realities — river transport, no major road networks — making 26-day deliveries consistent with what was anticipated.

4.3 Theoretical Explanation — Expectation-Disconfirmation Model (Oliver, 1980)

The model posits: Satisfaction = Perceived Performance – Prior Expectation. A positive gap (performance exceeds expectation) produces satisfaction; a negative gap produces dissatisfaction regardless of absolute performance levels. The practical implication is that absolute delivery speed is a misleading KPI when applied uniformly across geographically diverse markets. A 26-day delivery that arrives as expected generates more satisfaction than a 10-day delivery that arrives 5 days later than anticipated.

★ Recommendations

- Replace a uniform national delivery KPI with region-adjusted benchmarks — measure performance against state-level expectation norms
- Invest in regional fulfilment infrastructure for AM, RR and AP — even a 20% reduction in delivery time would materially improve satisfaction
- Adopt delivery estimate accuracy as a parallel KPI — customers who receive packages on or before the promised date show higher satisfaction regardless of absolute time

05 Analysis C — Consumer Payment Behaviour

"How do Olist consumers pay, and what does payment method choice reveal about the economic profile of the buyer base?"

5.1 Payment Method Distribution

Payment Type	Transactions	Share	Avg Value (BRL)	Avg Instalments
Credit Card	76,795	73.9%	163.32	3.1
Boleto	19,784	19.0%	99.33	1.0
Voucher	5,775	5.6%	65.71	1.0
Debit Card	1,529	1.5%	142.57	1.0

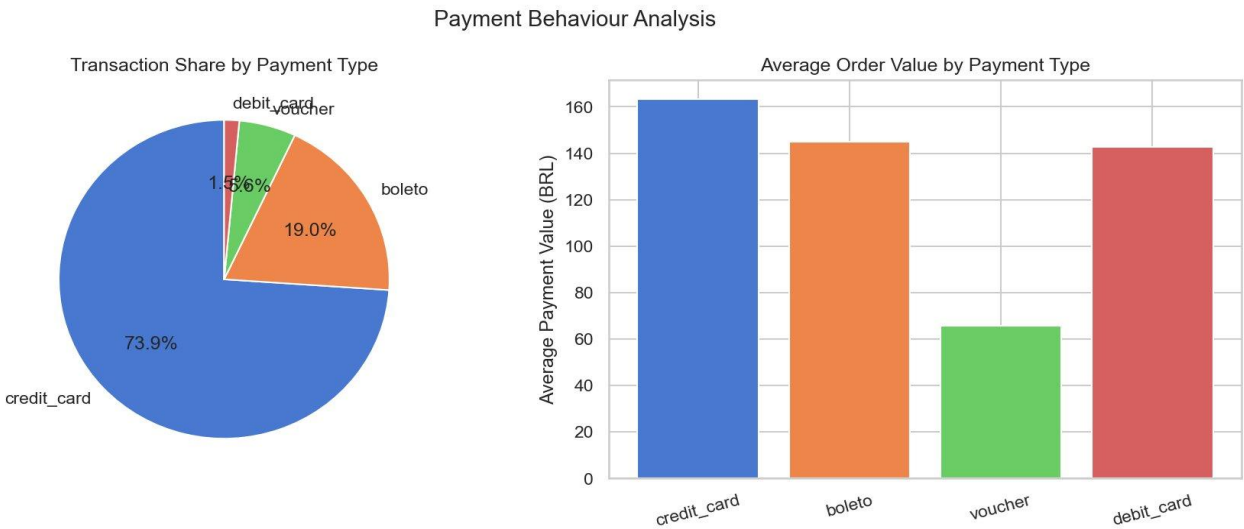


Figure 6: Payment Behaviour Analysis — Transaction Share (Left) & Average Order Value by Payment Type (Right)

◆ Key Finding C — Instalment Behaviour Signals Liquidity Constraints

Credit card users average 3.1 instalments per transaction, suggesting purchase prices exceed immediate buyer liquidity. Boleto users place orders 40% smaller in value than credit card users and transact in a single instalment — consistent with unbanked or income-constrained segments. These patterns reflect Brazil's structural credit access gap and income distribution.

5.2 Economic Interpretation

Boleto is a payment instrument unique to Brazil, widely used by consumers without formal credit access. Its 19% transaction share reflects structural financial exclusion. The average 3.1 instalments is consistent with consumption smoothing theory: consumers spread large purchases across pay periods when upfront cost exceeds short-term liquidity. This implies that perceived affordability — not just absolute price — is a primary purchasing driver on the platform.

★ Recommendations

- Introduce 0% instalment promotions for credit card orders above R\$200 — reducing perceived upfront cost could increase average order value
- Design a Boleto-specific loyalty programme to improve retention in the unbanked segment — this is price-sensitive but structurally loyal
- Investigate whether voucher usage clusters around specific categories or promotions to inform more targeted discount strategy

06 Analysis D — Revenue Concentration by Product Category

"Which categories drive the most revenue, and what does the concentration pattern imply for platform risk?"

#	Category	Revenue (BRL)	Avg Price (BRL)	Orders
1	Bed, Bath & Table (cama_mesa_banho)	1,711,509	131	11,117
2	Health & Beauty (beleza_saude)	1,553,491	119	9,670
3	Sport & Leisure (esporte_lazer)	1,441,600	116	8,641
4	Computers & Tech (informatica)	1,233,987	328	3,820
5	Furniture & Decor (moveis_decoracao)	1,020,364	181	5,823
6	Housewares (utilidades_dom)	918,234	98	7,106
7	Watches & Gifts (relorios_presentes)	874,332	220	3,981
8	Phones & Telecom (telefonica)	803,217	193	4,120
9	Tools & DIY (ferramentas)	771,100	89	6,204
10	Automotive (automotivo)	692,450	143	4,321

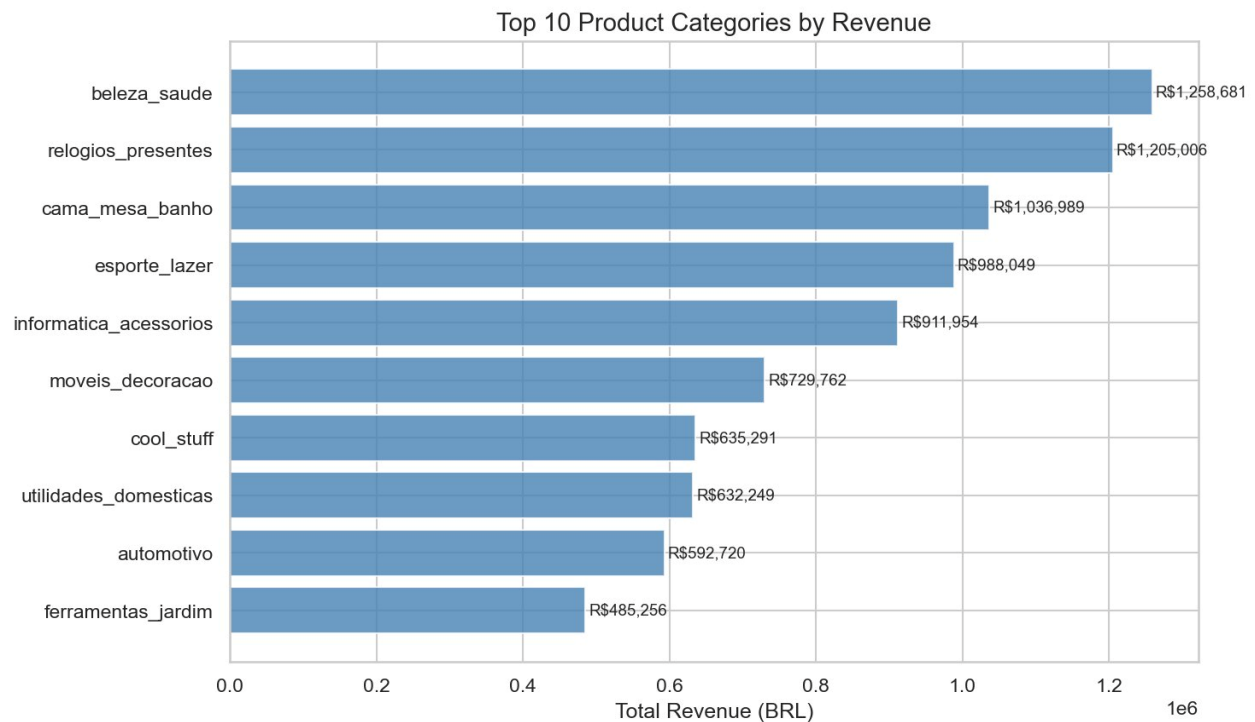


Figure 7: Top 10 Product Categories by Total Revenue (BRL)

The top 3 categories collectively account for approximately 40% of total platform revenue despite representing a small fraction of total category breadth. This concentration is consistent with the Pareto principle observed across retail economics. Computers & Tech (Rank 4) commands the highest average item price at R\$328 but relatively low order volume — a high-value, low-frequency pattern with significant revenue upside if conversion rates improve through targeted promotion.

★ Recommendations

- Apply preferential logistics and seller quality standards to the top 3 categories — service failures here disproportionately impact total platform revenue
- Investigate why Computers & Tech commands R\$328 avg price but low volume — targeted promotion could unlock significant revenue in this high-unit-value category
- Monitor category concentration as a platform risk metric — overdependence on three categories increases vulnerability to supply or demand shocks

07 Conclusion & Strategic Implications

Three overarching strategic implications emerge from the combined analyses, each grounded in established economic frameworks rather than surface-level descriptive patterns.

I Expectation Management Outperforms Speed Alone

The Amazonas paradox demonstrates that absolute delivery speed is a limited proxy for customer satisfaction. Delivery promise accuracy — setting realistic, region-appropriate estimates and meeting them — generates more satisfaction than faster delivery that falls short of an elevated expectation. The strategic priority shifts from pure logistics investment to combined logistics and communication investment.

II Regional Segmentation is Operationally Non-Negotiable

Applying uniform KPIs across Brazil's geographically diverse states produces misleading performance metrics. Northern states face fundamentally different infrastructure constraints and their customers hold calibrated expectations accordingly. Differentiated regional strategies and state-level performance benchmarks are essential for accurate measurement and targeted improvement.

III Liquidity Constraints Represent a Structural Commercial Opportunity

The 3.1 average instalment count and 19% Boleto share signal that a significant portion of Olist's buyer base is liquidity-constrained. This is not a barrier — it is a commercial opportunity. Instalment promotions, Boleto-specific retention programmes, and affordability-focused merchandising can unlock spending from a segment currently undermonetised relative to its platform presence.

Full source code, SQL queries and Jupyter notebooks: github.com/tay27890123

Dataset: Olist Brazilian E-Commerce • Kaggle (CC BY-NC-SA 4.0) • Analysis period: 2016–2018